SIGMIDI Foot Controller v0.1



SIGMIDI is a programmable TRS, USB and BLE MIDI foot controller built around the Arduino Nano ESP32 (ESP32-S3, 16 MB flash) and Waveshare 24382 color LCD. It sends PC/CC messages to its two buffered 3.5 mm TRS MIDI OUT sockets and mirrors them to native USB MIDI and BLE MIDI when those links are ready; it also provides encoder menus and a Wi-Fi configuration editor.

Features

- A four-switch base with up to three two-switch extensions: 4, 6, 8 or 10 momentary switches, detected individually through one resistor ladder.
- Independent ordered press and release lists of PC and CC messages, with channels 1–16 and a delay per message. PC supports STATIC, INCR and DECR with independent persistent counters. Added messages default to 5 ms. Per-switch Toggle mode alternates the two lists on successive presses.
- Two pedal inputs, independently selectable as two-wire resistors or three-wire potentiometers, with maximum resistance up to 10 kΩ, calibration, linear/log response and CC output.
- A 280 × 240 landscape LCD and a 20-detent encoder with click for local setup.
- An offline Wi-Fi editor for configuration, calibration, MIDI tests and JSON backups.
- Optional Pololu 2535 indicators, one per switch: white for a performed press action and yellow for release in momentary mode; white/off in Toggle mode.
- Two buffered 3.5 mm TRS MIDI OUT sockets (Type A), native class-compliant USB MIDI and BLE MIDI output. USB power or an optional 9 V center-negative input to Nano VIN; TRS MIDI OUT is powered from a diode-OR of USB VBUS and the converter 5 V rail.

Enclosures and switch identity

Left is the preferred enclosure and firmware default. Its display and encoder are on the left, with modular extensions added on the right. Left labels read **A B C D E / 1 2 3 4 5** across the upper/lower rows. Right has the electronics on the right and reverses each row.

Select **Settings** → **Enclosure variant** in the Wi-Fi editor or **EDIT CONFIG** → **ENCLOSURE** on the LCD. Left combines 180° LCD rotation with Left list order; Right uses 0°. MIDI actions, names, LED assignments and calibration belong to permanent electrical taps, independently of the selected variant or installed switch count.

The main enclosure measures **350 × 92 × 54 mm**, with a 30 mm raised row, 65 mm horizontal pitch, 32.5 mm stagger and 53 mm row spacing. The Left electronics shell has a 25 mm-wide icon raised 0.6 mm between A and B. Main and modular assemblies are available for both variants.

The Nano mounts component-side up directly on four reinforced posts in each electronics base. See the direct Nano mounting guide. Use the Nano pin/power map, electrical diagrams and editable KiCad schematic.

Install and configure

Use the preserved release bundles. From the project root:

```
python3 scripts/serve_flasher.py
```

Open **http://localhost:8766/** in desktop Chrome or Edge, connect the Nano over USB-C in ROM download mode (jumper B1 to GND, tap RST, remove the jumper) and choose **Connect & install**. Leave **Erase device** unchecked to keep compatible saved settings. The flasher serves the complete local installer from `distribution/web-flasher/`.

On the controller, enter **Wi-Fi Setup**, join the displayed `SIGMIDI-xxxx` network, and open **http://192.168.4.1/**. Choose the enclosure, ladder wiring and installed switch count, calibrate the inputs, configure MIDI actions and save. Normal performance runs with Wi-Fi off. The BLE name is **SIGMIDI Foot Controller v0.1**.

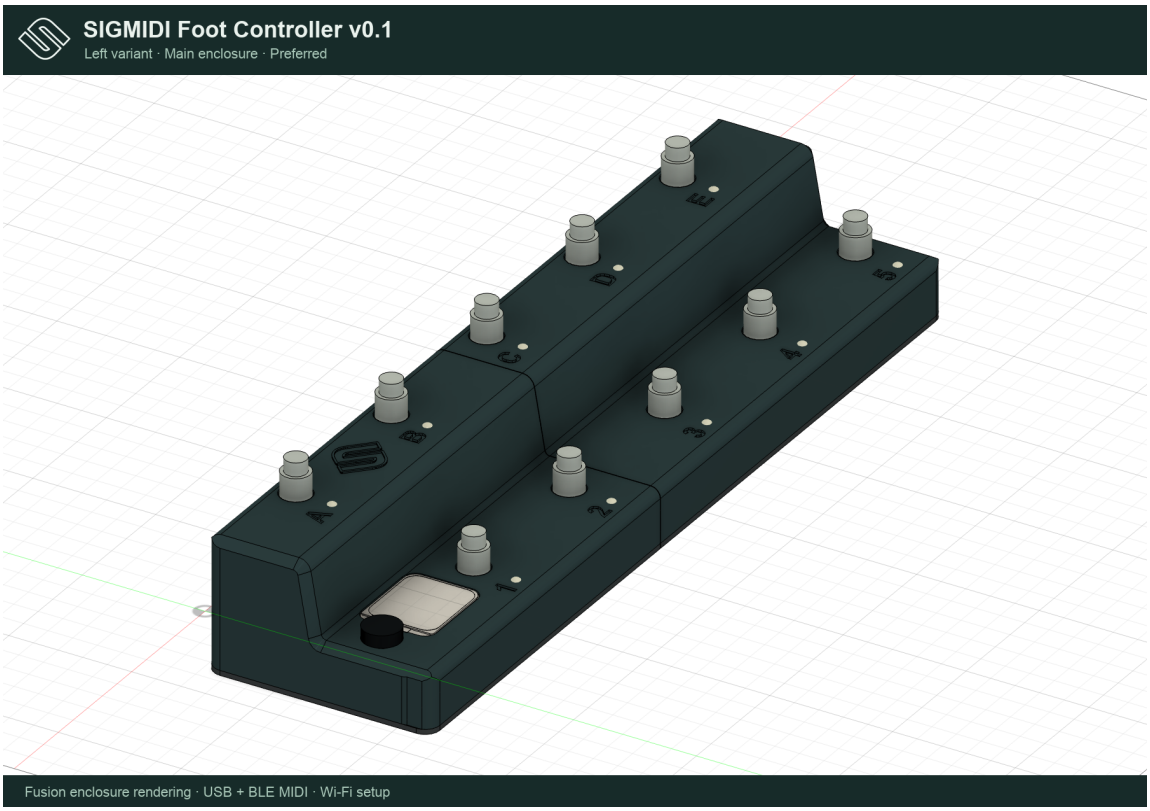
Documentation and files

Area	Guide
Step-by-step assembly, wiring and first power-up	Assembly guide PDF · Editable source
USB/BLE connections, performance and editing; HX Stomp example in Appendix A	User manual PDF · Editable source
Parts, hardware and enclosure quantities	Bill of materials PDF · Shopping CSV · Document source
Every other documentation page as a printable PDF	Reference PDFs · Rebuild with `scripts/build_reference_pdfs.py`
Electrical wiring and GPIO	Hardware
Editable electrical schematic	KiCad project
Modular taps and resistor values	Resistor ladder
Menus, configuration and MIDI behavior	Firmware
Building and commissioning	Bring-up
Browser installation and hosting	Web flasher
LED wiring and configuration	LEDs
Power wiring and load budget	Power
CAD and manufacturing files	Design
Arduino Nano ESP32 components-up mounting	Direct base mounting
Screenshots and enclosure renders	Previews

The complete Fusion model contains the native component hierarchy for all four assemblies and the modular part libraries. Firmware source is in `src/`, the embedded editor in `web/`, and tests in `tests/`. Release binaries and installers are in `distribution/`; `build/` contains disposable compiler outputs when a build is run.

Validation status

The firmware targets ESP-IDF 5.3.1 on ESP32-S3. Host tests cover configuration, MIDI scheduling, UI navigation, display mapping, switch identity and LED feedback. CAD, mesh and slicer checks accompany the manufacturing files. Physical USB flashing, iPad interoperability, printed fit/load strength and sustained thermal operation require validation on the assembled hardware.



SIGMIDI preferred Left enclosure

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